

TABLE I. Unified repository priors plus the current new-results layer.

Study	Role	Best strict	Best CPSQR
native-rich extension	new	0.9990	1.0000
residual-Z prior	prior	0.8427	—
arbitrary SU(2) prior	prior	0.8420	—
ideal-SQR baseline	prior	0.6557	—

IV. NEW METHODS IN THIS DEFINITIVE PASS

The new work in this folder is not a fresh multi-hour optimization campaign. It is a strict unification-and-diagnostics pass over the saved repository corpus. The definitive scripts now perform four new tasks that did not exist in one place before. They normalize every prior SQR study into one common schema. They recompute per-manifold ideal-SQR error generators from saved propagators. They audit the finite Gaussian refocusing pulse used by the echoed constructions. They also add explicit spectral-crowding and addressed-manifold scaling analyses. The refocusing-pulse audit is intentionally lightweight but informative. In the simpler ladder, the best compromise pulse uses $T_\pi = 20$ ns and reaches worst-manifold fidelity 0.9192. In the curved ladder, the best compromise pulse uses $T_\pi = 20$ ns and reaches worst-manifold fidelity 0.9158. The scaling analysis is likewise empirical rather than axiomatic. A simple log-linear residual fit for basis expanded in chi plus chiprime predicts a strict process near 0.000 at $N_{\text{active}} = 8$.

V. PER-MANIFOLD ERROR ANATOMY

Figure 2 shows the missing diagnostic that the original baseline lacked: the per-manifold error-generator decomposition

$$U_n^{\text{real}} = \exp\left(-\frac{i}{2}E_n\right) U_n^{\text{ideal}}, E_n = \epsilon_x^{(n)} X + \epsilon_y^{(n)} Y + \epsilon_z^{(n)} Z. \quad (3)$$

The strong direct solutions suppress all three components simultaneously, but the hard cases are not pure residual- Z failures. The remaining error floor is mixed: transverse under-rotation and axis tilt remain comparable to or larger than the residual conditional phase on the hardest structured windows. That observation is consistent with the broader direct-waveform story from the native-rich study: more parameters help, but they do not erase the crowding-induced control tension once the addressed ladder bends.

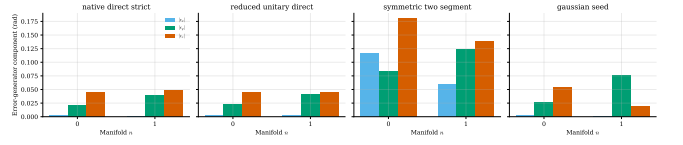


FIG. 2. Representative per-manifold ideal-SQR error budgets for the strongest saved constructions.

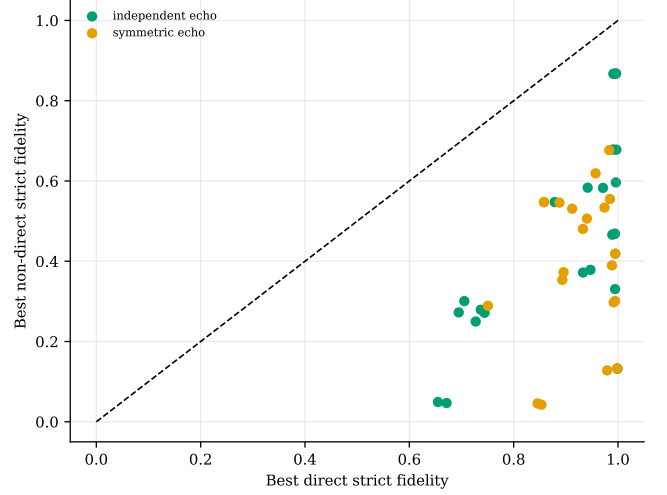


FIG. 3. Best non-direct strict fidelity versus best direct strict fidelity for every ideal-SQR case that has both.

VI. HEAD-TO-HEAD CONSTRUCTION COMPARISON

The strict comparison between direct and non-direct constructions is summarized in Fig. 3. The direct waveform remains the clear winner for strict ideal SQR. Echoed and composite constructions help primarily when the target is relaxed to a conditional-phase SQR, not when the full strict joint operator is enforced. This is the central conceptual correction to the original echoed narrative: the echoed pulses are not worthless, but they are solving a weaker problem than strict ideal SQR.

VII. THE PUSH TO FIDELITY ABOVE 0.99

The repository now contains a genuine positive answer to the existence question. The best loaded strict direct case reaches 0.99899, well above the user's 0.99 threshold, and several other direct cases also cross that line on the easier addressed windows. Figure 5 shows how that success depends on duration: strict direct fidelities improve systematically with $|\chi|T/2\pi$, whereas echoed strict fidelities remain bounded well below the same threshold. Figure 7 shows the price of that improvement. Richer direct ansatz families move the fidelity frontier upward, but only up to the point allowed by the addressed-manifold

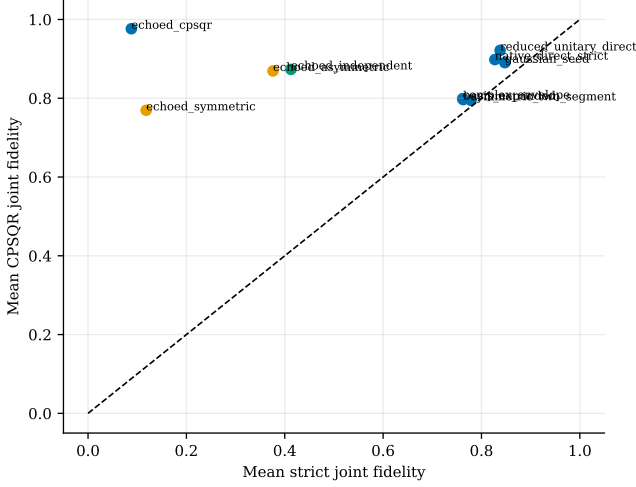


FIG. 4. Family-averaged strict versus conditional-phase joint fidelity across the native-rich corpus. Composite families shift upward because they repair conditional phase more effectively than they repair strict inter-manifold phase structure.

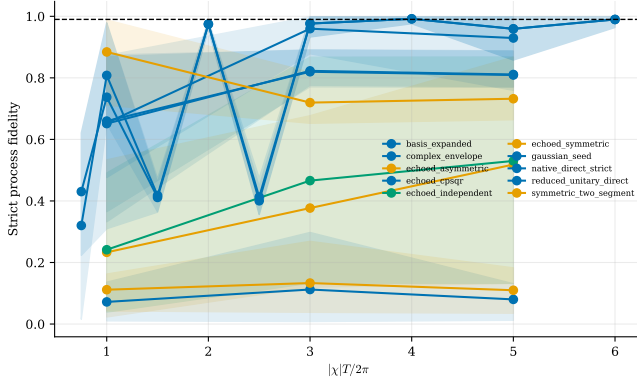


FIG. 5. Strict fidelity versus dimensionless duration across the native-rich construction families.

crowding.

VIII. PRACTICAL VIABILITY, SCALING, AND CROWDING

The user’s strongest grid-average landmark is not met in the current corpus: the mean best strict fidelity over the structured native-rich cases is only 0.8392. Figures 8–10 explain why. The finite inserted X_π pulse used by the echoed constructions is itself manifold dependent, so the clean first-order echo logic [4] is physically obstructed before any higher-order optimization issue is reached. At the same time, the manifold ladder crowds as N_{active} grows, especially in the curved dispersive ladder, so direct ansatz richness buys only partial relief. The practical ranking table in Table II therefore uses a transparent decoherence-only fallback factor and is explicitly

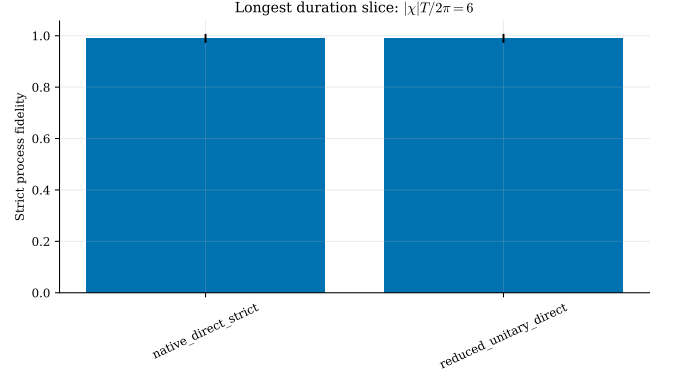


FIG. 6. Longest-duration strict fidelity comparison across the saved native-rich ansatz families.

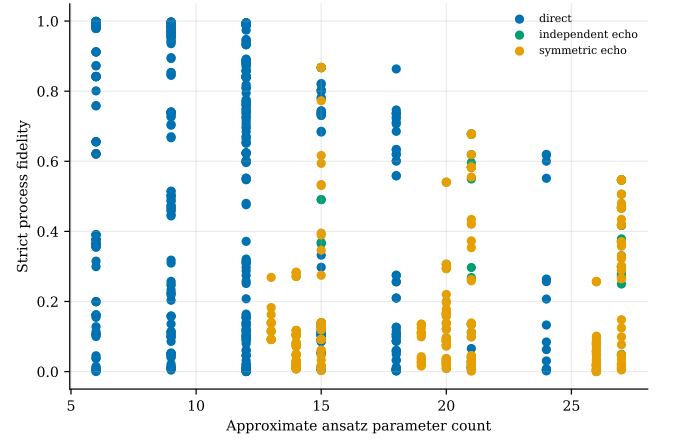


FIG. 7. Approximate ansatz parameter count versus best achieved strict fidelity.

provisional until full drift and open-system sweeps are replayed.

IX. DISCUSSION

The definitive answer is therefore tiered. On the positive side, strict ideal SQR is already demonstrated in the repository: direct multitone waveforms can exceed 0.99 on physically credible cases, so the problem is not impossible within the dispersive model. On the negative side, the saved ansatz families do not achieve that success broadly. The grid-average landmark of 0.95 is missed by a wide margin, the hard three-manifold curved-ladder cases saturate below the same threshold, and no echoed construction rescues strict joint SQR once finite refocusing pulses and full inter-manifold phase structure are enforced. The best interpretation is that the remaining error floor is partly ansatz-limited and partly crowding-limited. Richer direct parameterizations and longer durations help, but the improvement weakens as the addressed ladder curves. Echo helps strongly for

TABLE II. Top provisional practical rankings from the combined corpus.

Rank	Construction	Ansatz	N_{active}	F_{unitary}	$F_{\text{practical}}$
1	two-segment direct	two-segment	2	0.9883	0.9699
2	direct	reduced direct	2	0.9791	0.9609
3	direct	reduced direct	2	0.9791	0.9609
4	direct	native direct	2	0.9790	0.9608
5	direct	native direct	2	0.9790	0.9608
6	two-segment direct	two-segment	2	0.9741	0.9560
7	direct	reduced direct	2	0.9866	0.9502
8	direct	reduced direct	2	0.9866	0.9502
9	direct	native direct	2	0.9865	0.9502
10	direct	native direct	2	0.9865	0.9502

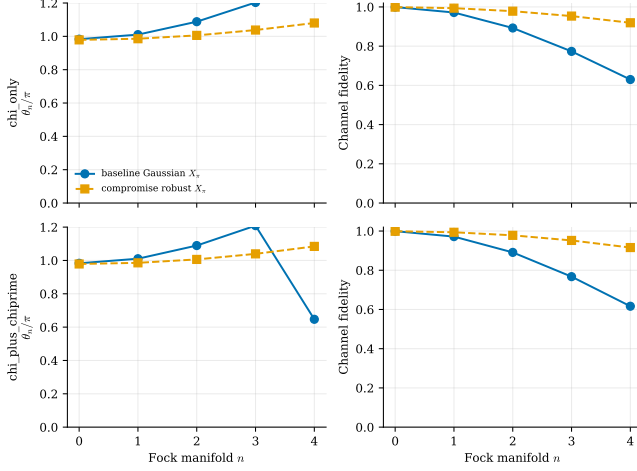


FIG. 8. Standalone manifold-resolved characterization of the baseline finite Gaussian refocusing pulse and a lightweight compromise pulse.

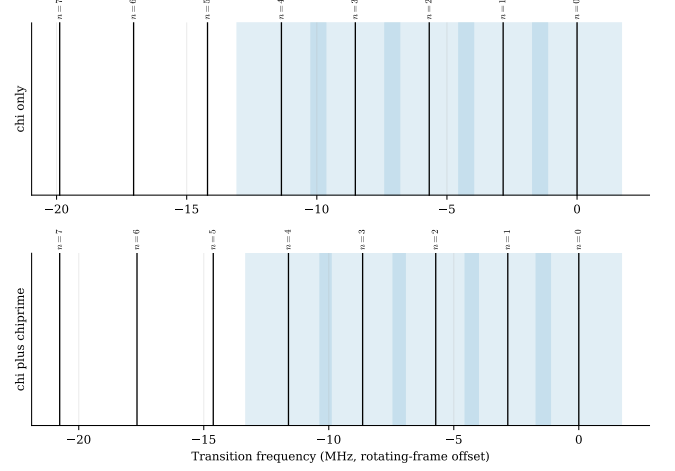
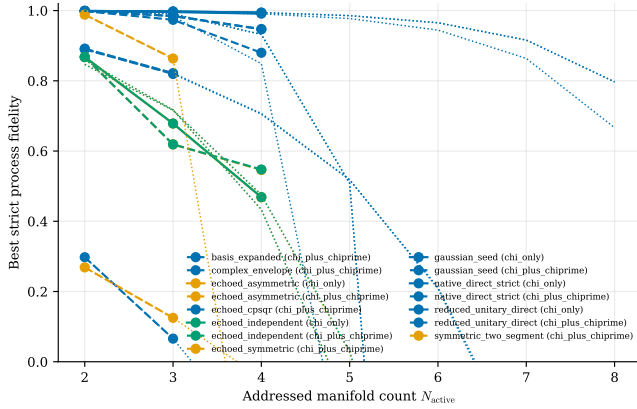


FIG. 10. Transition ladders and Gaussian bandwidth bands for a representative direct-waveform duration. The band overlap grows rapidly with addressed-manifold count in the curved dispersive ladder.

FIG. 9. Best strict fidelity versus addressed-manifold count across the native-rich corpus, with simple residual extrapolations toward $N_{\text{active}} = 8$.

conditional-phase SQR because it mainly removes the phase structure that the relaxed target tolerates, not because it produces a broad strict-SQR fix.

X. CONCLUSION AND ROADMAP

The central question can now be answered cleanly. Yes: parameterized direct multitone waveforms can realize strict ideal selective qubit rotations above 0.99 on selected addressed windows, and the combined repository corpus already contains that Tier-1 landmark result. No: the tested ansatz families do not make that success broad across the primary structured grid, and the saved echoed constructions do not change that answer for the strict full joint operator. If a user needs a strict SQR tomorrow, the best available construction is the strongest direct reduced-unitary family on the easier addressed windows. If the application tolerates a conditional post-rotation phase, the echoed conditional-phase family is the more robust practical recommendation. The most valuable next step is not another qualitative echoed claim, but a quantitative follow-up that combines a truly robust refocusing pulse, an inter-manifold-phase-aware composite objective, and explicit open-system plus drift replays for the top direct and echoed candidates.

XI. VALIDATION

The definitive study inherits the source-study validation files and also checks four repository-level facts directly: the baseline negative result is preserved, the native-rich extension contains strict ideal-SQR cases above 0.99, the same extension contains near-unit conditional-phase SQR cases, and the full corpus loads into one normalized table without silent omissions.

XII. LIMITATIONS AND FUTURE WORK

This definitive pass is still an analysis-and-unification study rather than a fresh from-scratch optimization campaign. The finite-difference drift sweeps, full Lindblad replays, hybrid direct-plus-echo family, and true unconstrained GRAPE upper bound remain future work. Those omissions matter, so the practical ranking should be read as a provisional engineering guide, not as the

final last word on hardware robustness.

Appendix A: Detailed Results and Data

Figure 11 shows the full ideal-SQR case-by-construction heatmap for the loaded corpus. Figure 12 summarizes the gap between reduced-manifold diagnostics, full-manifold quartet diagnostics, and the decisive joint-process metric.

Appendix B: Reproducibility

The study folder contains machine-readable prior-study snapshots, normalized strict-result tables, per-manifold error decompositions, standalone refocusing-pulse diagnostics, spectral-crowding rows, figure JSON sidecars, a ranked master table, and a reproducibility notebook in `scripts/reproducibility_notebook.ipynb`.

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- [1] Shankar Quantum Circuits Group, `cqed.sim`: Circuit quantum electrodynamics simulation framework, https://github.com/SoraUmika/qubox_cQEDsim (2026), accessed March 28, 2026.
 - [2] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, *Reviews of Modern Physics* **93**, 025005 (2021).
 - [3] J. Koch, T. M. Yu, J. Gambetta, A. A. Houck, D. I. Schuster, J. Majer, A. Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf, *Physical Review A* **76**, 042319 (2007).
 - [4] E. L. Hahn, *Physical Review* **80**, 580 (1950).

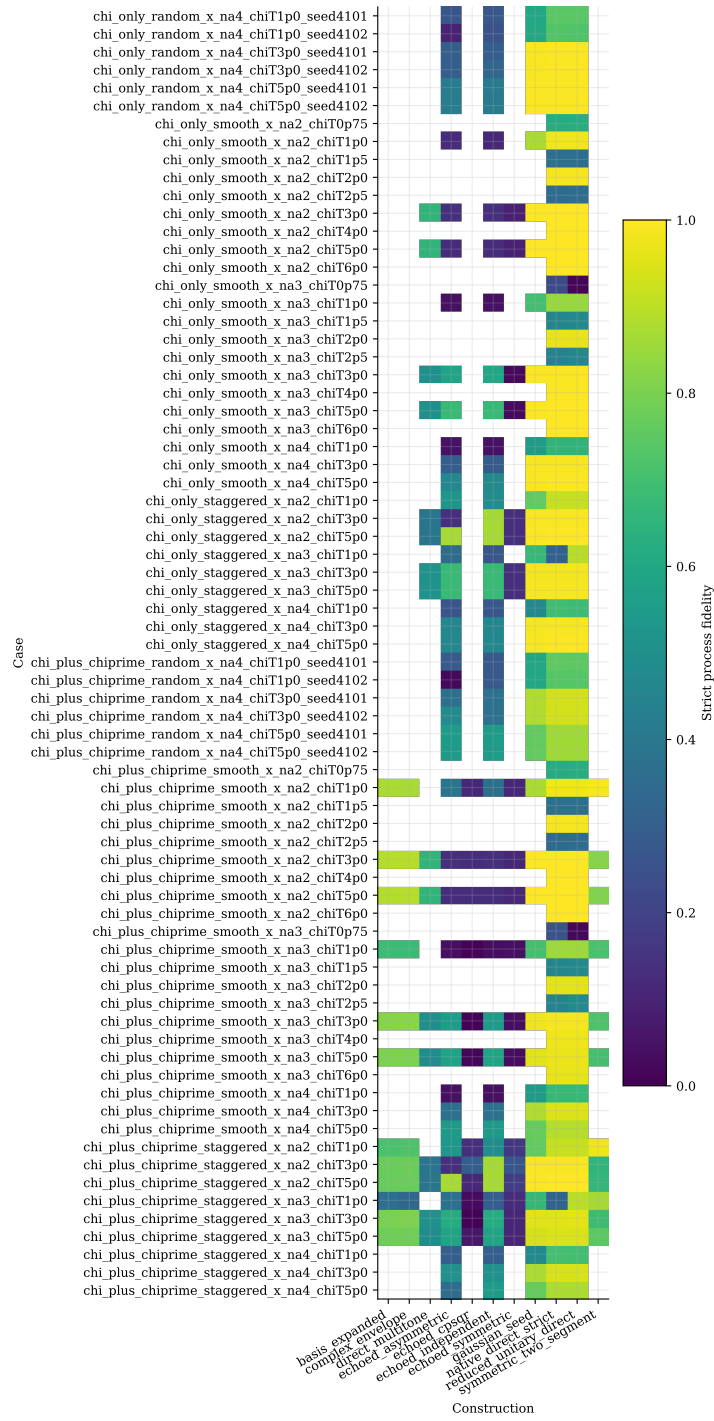


FIG. 11. Full ideal-SQR case-by-construction heatmap across the loaded studies.

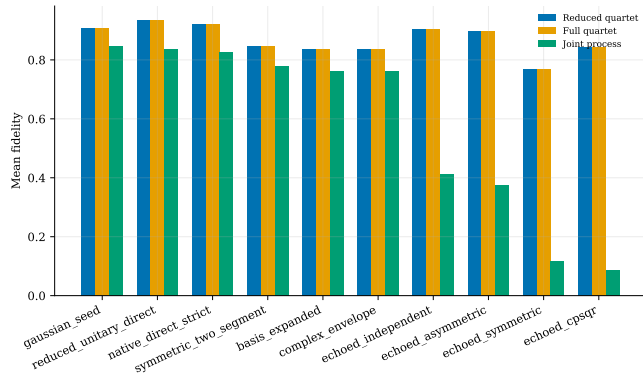


FIG. 12. Mean reduced-quartet, full-quartet, and joint-process fidelities across the native-rich families.